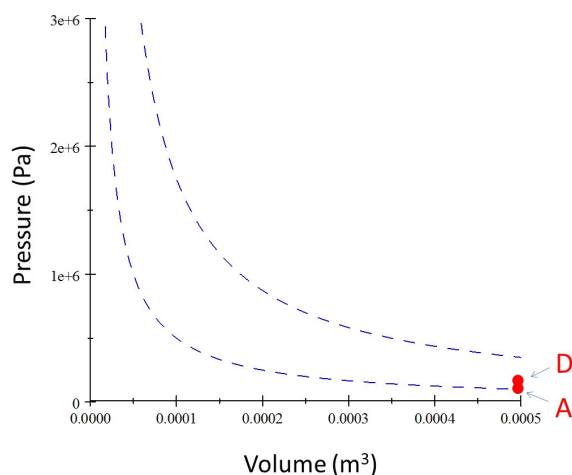


1.4 Process $D \rightarrow A$

Fill out the following table for process $D \rightarrow A$

State	T (K)	P (kPa)	V (m ³)	n (mol)	E _{int} (J)	—
D	445.28	151.97	0.000 5	2.0525×10^{-2}	189.96 J	—
A	293	100	0.000 5	2.0525×10^{-2}	125.00	—
Process	Q (J)	w _{int} (J)	ΔE _{int} (J)	w _{eng} (J)	Q _h	Q _c
DA						

Process $D \rightarrow A$ is an isochoric process. We are nearing the end of the cycle, In process $D \rightarrow A$ a valve opens letting the pressure drop quickly.



Since this happens quickly, the piston does not have time to change position, so the volume does not change. Process DA is (almost) isochoric. We already know P_A , V_A , n , and T_A because that is where we started and this is a cyclic process. But we need to find the process work, heat and change in internal energy. Since the volume does not change, $W_{DA} = 0$ and $\Delta E_{DA} = Q_{DA}$ which we can find using

$$\begin{aligned}
 Q_{DA} &= nC_V\Delta T \\
 &= n\frac{5}{2}R(T_A - T_D)
 \end{aligned}$$

which gives

$$\begin{aligned}
 Q_{DA} &= (2.0525 \times 10^{-2} \text{ mol}) \frac{5}{2} \left(8.314 \frac{\text{J}}{\text{mol K}} \right) (293 \text{ K} - 445.28 \text{ K}) \\
 &= -64.964 \text{ J}
 \end{aligned}$$

Which tells us that we have lost energy from the engine by heat. So Q_{DA} must contribute to Q_c .

State	T (K)	P (kPa)	V (m³)	n (mol)	E_{int} (J)	—
D	445.28	151.97	0.000 5	2.0525×10^{-2}	189.96 J	—
A	293	100	0.000 5	2.0525×10^{-2}	125.00	—
Process	Q (J)	w_{int} (J)	ΔE_{int} (J)	w_{eng} (J)	Q_h	Q_c
DA	-64.964	0	-64.964	0	0	64.964